

Subject Description Form

Subject Code	SFT6986
Subject Title	Guided Study in Comfort
Credit Value	3
Level	6
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims at providing students with a guided in-depth study on the fundamentals and practices of evaluating and improving comfort of fashion and textile products in terms of physical comfort, physiological comfort, psychological comfort, etc., with comprehensive understanding of the complex interactions between fabric and garment design, personal activities, environment and climate, and physical/physiological/psychological variables.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand basic concepts of comfort and the factors affecting comfort including clothing, human body, and environment. - Demonstrate critical awareness of the latest fashion technologies and innovations on comfort and integrate them into the process of fashion design and development. - Analyse and evaluate the comfort properties of apparel products and define the types of testing instruments. - Develop plans of comfort improvement for existing fashion products and latest clothing and wearable technologies and present own ideas and views effectively.
Subject Synopsis/ Indicative Syllabus	This subject will provide a guide for research students to carry out in-depth study in evaluating and improving comfort of textiles and clothing, including concepts and principles, comparison of the comfort performance of fashion and wearable products, objective and subjective measurement, latest technologies and innovations (e.g., adaptive comfort), case illustrations such as daily fashion wear, firefighters' suit, varying types of athletic gear, and protective ensembles for military, industrial, and medical personnel, assessment of mechanical, thermal, tactile, and wear comfort, and relationship between comfort and fabric structures/materials.
Teaching/Learning Methodology	Fundamental concepts and knowledge will be introduced through tutorials. Laboratories will be conducted on the typical topics such as evaluation of thermal and moisture comfort to help students integrate theory with practice. The subject lecturer will suggest reading materials based on student's research area and interest. Students are required to give an oral presentation and submit a written report on an assigned topic related to their PhD/MPhil project.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d		
	1. Coursework	40%	✓	✓	✓	✓		
	2. Examination	60%	✓	✓	✓	✓		
	Total	100 %						
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Coursework helps students demonstrate and extend their knowledge of comfort and formulate a solution for improving comfort of a fashion product in a broader perspective. The examination provides an opportunity for the students to demonstrate their critical thinking and awareness of comfort evaluation and innovations from a professional perspective in response to the changing needs of the fashion academy and industry.								
Student Study Effort Expected	Class contact:							
	▪ Tutorial/presentation/discussions						20 Hrs.	
	▪ Laboratory						6 Hrs.	
	Other student study effort:							
	▪ Literature studies						36 Hrs.	
	▪ Self-reading/assignments						60 Hrs.	
	Total student study effort						122 Hrs.	
Reading List and References	<u>Monographs:</u> 1. Song, G. (Ed.). (2011). <i>Improving comfort in clothing</i>. Elsevier. 2. Fan, J., & Hunter, L. (2009). <i>Engineering apparel fabrics and garments</i>. Elsevier. 3. Apurba, D., & Alagirusamy, R. (2010). <i>Science in clothing comfort</i>. Woodhead Publishing Limited. 4. Li, Y. (2001). The science of clothing comfort. <i>Textile progress</i>, 31(1-2), 1-135. 5. Pan, N., & Gibson, P. (Eds.). (2006). <i>Thermal and moisture transport in fibrous materials</i>. Woodhead Publishing. 6. Jamil, I. F. Z., & Bakar, B. A. (2021). Thermal Comfort Properties On Army Uniform With Objective And Subjective Evaluation. <i>Progress in Engineering Application and Technology</i>, 2(2), 887-892. 7. Gao, S., Ooka, R., & Oh, W. (2021). Experimental investigation of the effect of clothing insulation on thermal comfort indices. <i>Building and Environment</i>, 187, 107393. 8. Zhang, X., Chao, X., Lou, L., Fan, J., Chen, Q., Li, B., ... & Shou, D. (2021). Personal thermal management by thermally conductive composites: A review. <i>Composites Communications</i>, 23, 100595. 9. Awais, M., Krzywinski, S., & Wendt, E. (2021). A novel modeling and simulation approach for the prediction of human thermophysiological comfort. <i>Textile Research Journal</i>, 91(5-6), 691-705. 10. Wilfling, J., Havenith, G., Raccuglia, M., & Hodder, S. (2021). Consumer expectations and perception of clothing comfort in sports and exercise garments. <i>Research Journal of Textile and Apparel</i>. 11. Tadesse, M. G., Loghin, C., Dulgheriu, I., & Loghin, E. (2021). Comfort Evaluation of Wearable Functional Textiles. <i>Materials</i>, 14(21), 6466.							

	12. Kaplan, S., & Okur, A. (2008). The meaning and importance of clothing comfort: A case study for Turkey. <i>Journal of Sensory Studies</i>, 23(5), 688-706. 13. Havenith, G., Holmér, I., & Parsons, K. (2002). Personal factors in thermal comfort assessment: clothing properties and metabolic heat production. <i>Energy and buildings</i>, 34(6), 581-591. 14. Li, Y. (2005). Perceptions of temperature, moisture and comfort in clothing during environmental transients. <i>Ergonomics</i>, 48(3), 234-248. 15. Kamalha, E., Zeng, Y., Mwasiagi, J. I., & Kyatuheire, S. (2013). The comfort dimension; a review of perception in clothing. <i>Journal of sensory studies</i>, 28(6), 423-444. <u>Journals:</u> 1. Science 2. Nature 3. Science Advances 4. Nature Communications 5. Advanced Materials 6. Advanced Functional Materials 7. The Journal of The Textile Institute 8. Textile Research Journal 9. International Journal of Heat and Mass Transfer 10. Ergonomics 11. Applied Ergonomics 12. Measurement Science and Technology 13. Building and Environment
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